GENERATIVE AI MODULE · FINAL PROJECT

Clinical Research Compliance Assistant

An expert agent on ICH GCP, expedited safety reporting and the Declaration of Helsinki — built with Gemini, ChromaDB and LangGraph, with memory and verbatim citations.

David Naranjo Ramírez Master in Data Science & AI · Evolve Academy · November 2026

The question a CRA asks at 17:50 on a Friday

AN SAE JUST CAME IN

“Is this a SUSAR? Who reports it, to whom, and by when?”

The answer is split across three texts — ICH E2A (timelines), ICH E6(R3) (investigator and sponsor duties) and the Declaration of Helsinki (ethics) — and it has to be quoted exactly, not paraphrased.

Three authoritative sources, ~90 pages

ICH E6(R3) GCP (2025) · ICH E2A · WMA Declaration of Helsinki (2024)

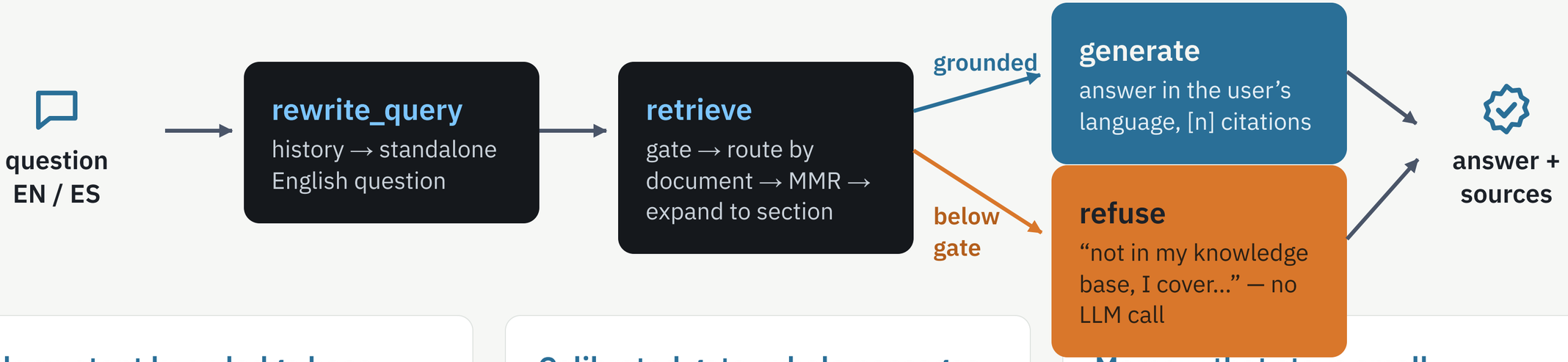
Traceability is the requirement

Every claim cited to document, section and page; refuse what the sources do not cover

Why me

5 years supporting clinical-trial software (Medidata Rave, Pfizer, Novo Nordisk) · ICH GCP certified · previous project: FAERS GLP-1 pharmacovigilance

An explicit graph: the knowledge base is used on every query



Idempotent knowledge base

801 chunks, gemini-embedding-2. A versioned embedding cache + manifest: nothing is re-embedded, and Chroma rebuilds from it with zero API calls — on Streamlit Cloud too.

Calibrated gate, whole passages

Threshold 0.69 read off in-domain vs out-of-domain queries, not guessed. Then a metadata filter for documents the question names, and each chunk widened to its section so lists arrive whole.

Memory that stays small

LangGraph checkpointer per thread_id; the last 10 messages reach the model; retrieved passages live in state, not in the history.

Stack: Gemini 3.5 Flash-Lite · gemini-embedding-2 · ChromaDB · LangGraph · Streamlit — github.com/dnarram/clinical-research-compliance-assistant

The system prompt: each version answers an observed failure

V1 · CLASS BASELINE

“Answer from the documents”

- Correct and fluent, but no citation markers: no sentence can be traced to a passage
- No rule for missing information, no scope, no language policy

V3 · CITATIONS AND LANGUAGE

[n] on every sentence, sources attributed, answer in the user’s language

- The golden-set run found: a multi-step question refused outright
- Its Spanish refusal written in English
- Truncated lists presented as complete

V4 · IN USE

Graded refusals, multi-step answers, complete lists

- Answer the covered part, name the uncovered part; refuse only when nothing is relevant
- Build chains step by step, citing each step
- Give every list item, or say the list continues — in the user’s language

Hard constraints in code, judgement in the prompt. The answer language is a deterministic line in the user message, out-of-domain questions never reach the model, and truncated lists were fixed in retrieval (section expansion), not with more prompt.

Full prompt and rationale: README · docs/prompt_iterations.md (v2 was a design step, not evaluated on its own)

Live demo: four questions, four behaviours

1 · GROUNDED ANSWER WITH CITATIONS

What is the timeline for expedited reporting of a fatal unexpected ADR?

Expect: 7 calendar days + 8 for the complete report, quoted verbatim — ICH E2A §1, p.7. Open the sources panel.

2 · MEMORY + LANGUAGE POLICY

¿Y para las demás reacciones graves e inesperadas?

Expect: rewritten into a standalone English question (shown as “Searched as”), answered in Spanish: 15 calendar days.

3 · TWO SOURCES, ATTRIBUTED SEPARATELY

¿Qué exigen ICH E6(R3) y la Declaración de Helsinki para incluir a participantes que no pueden dar su consentimiento?

Expect: both documents named, so the top-6 is shared between them; E6(R3) Principle 2.1 / §2.8.13 vs Helsinki ¶28 contrasted, each with its own citation.

4 · REFUSAL, NO MODEL CALL

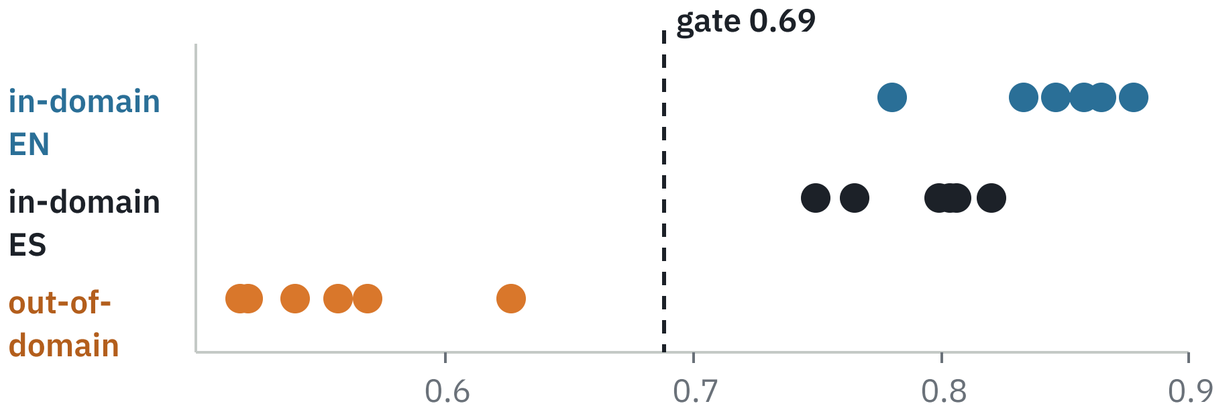
¿Qué fármaco GLP-1 es mejor para perder peso?

Expect: best relevance ≈0.64, below the 0.69 gate — “not in my knowledge base; I cover…”. The answering model never saw the question.

Then one node-by-node trace in the notebook (rewrite → retrieve → generate) · Fallback: local Streamlit, executed notebook

Evaluation: judged correctness 0.78 → 0.95, confirmed on a held-out set

Best relevance score per query (top-3, cosine) — 6 EN/ES pairs + 6 out-of-domain



Spanish queries retrieve the same passages ~0.04 lower; lowest in-domain 0.75 vs highest out-of-domain 0.63. The gate is the midpoint of that gap and is recorded with the embedding model.

GOLDEN SET

Golden: 25 items × EN/ES. Held-out: 12 new items, written after v4 was frozen, run once. LLM judge vs expert references.

Same scoring	v3	v4 + fixes
Judge, golden	0.78	**0.95**
Judge, held-out	0.95	**1.00**
Answer keywords	0.82	**0.97**
Section hit EN / ES	0.95 / 1.0	0.95 / 0.95
Off-topic refused	100%	100%
Honest, not indexed	100%	100%

scripts/02_sanity_retrieval.py --calibrate · scripts/03_evaluate.py --with-answers · docs/evaluation.md

What it does not do yet, and what comes next

⚠ Limitations

- Three documents by design: EU CTR, GVP modules and FDA rules are refused or answered only from what E6(R3) says
- Multi-hop questions: one retrieval pass misses a hop — the reporting chain still lacks the 15-day timeline
- The refusal gate is corpus- and model-specific: change either, recalibrate
- Memory lives in the process; a restart forgets the threads
- Free tier: 1,000 embedded texts per day — the corpus build used 800 of them once, so query vectors are cached too

Next steps, in order of value

1. Query decomposition: one retrieval per step of a multi-hop question
2. EMA GVP Module VI and EU CTR 536/2014 — ingestion is per document and idempotent
3. Faithfulness judge: answer vs. retrieved passages (today's judge checks correctness vs. expert references)
4. Hybrid retrieval (BM25 + vectors) for exact section numbers and acronyms
5. A guard_input node (prompt injection, PII) and SqliteSaver for persistent threads

Thank you — questions?

github.com/dnarram/clinical-research-compliance-assistant · live app on Streamlit Cloud · regulatory information, not legal or medical advice